WORKED SOLUTION

STEP II 1992, Question 4

Pure • Axioms for Differentiation

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q4

This question has a full guided treatment in *Beyond STEP*, Section 6.5, *Differentiation from Axioms*: https://beyondstep.keithmcnulty.org/ch06/derivations.html

The key idea

The four rules are exactly the familiar properties of differentiation: the product rule, linearity and $\frac{\mathrm{d}}{\mathrm{d}x}x = 1$. The question asks us to show that these properties alone pin the operation down completely on polynomials. The trick for constants is to feed the product rule a product whose answer we already know, such as $1 \times 1 = 1$. After that, the product rule builds up every power of x one step at a time, and linearity handles the rest.

Throughout, we must use *only* the four given rules. We may not assume anything about $\Diamond$ just because it looks like differentiation; that is what we are trying to prove.

Solution

Constants

? **Ask yourself** We know nothing about $\Diamond 1$. How can the rules tell us what it is?

Apply the product rule (i) to the polynomials $\mathrm{f}(x) = \mathrm{g}(x) = 1$. Their product is 1 again, so

$$\Diamond 1 = \Diamond(1 \cdot 1) = 1 \cdot \Diamond 1 + 1 \cdot \Diamond 1 = 2\,\Diamond 1.$$

Subtracting $\Diamond 1$ from both sides gives $\Diamond 1 = 0$, the zero polynomial.

Now let $\mathrm{f}(x) = c$ be any constant polynomial. Then $\mathrm{f}(x) = c \cdot 1$, and c is a constant, so rule (iv) gives

$$\Diamond c = \Diamond(c \cdot 1) = c\,\Diamond 1 = c \cdot 0 = 0.$$

This covers $c = 0$ too, so $\Diamond \mathrm{f}(x) = 0$ for every constant polynomial, as required.

(Another route to $\Diamond 1 = 0$: write $x = x \cdot 1$, so rule (i) and rule (iii) give $1 = 1 \cdot 1 + x\,\Diamond 1$, hence $x\,\Diamond 1 = 0$, and a polynomial that gives 0 when multiplied by x must be the zero polynomial.)

The powers x^2 and x^3

Using the product rule with $\mathrm{f}(x) = \mathrm{g}(x) = x$, and then rule (iii),

$$\Diamond x^2 = \Diamond(x \cdot x) = x\,\Diamond x + x\,\Diamond x = x + x = 2x.$$

Next, $x^3 = x^2 \cdot x$, so by the product rule, rule (iii) and the result we have just found,

$$\Diamond x^3 = x\,\Diamond x^2 + x^2\,\Diamond x = x \cdot 2x + x^2 \cdot 1 = 3x^2.$$

Any polynomial

Step 1: powers of x. We prove by induction that

$$\Diamond x^n = n x^{n-1} \qquad \text{for every integer } n \geq 1.$$

For $n = 1$ this is rule (iii). Suppose it holds for some $n = k \geq 1$. Writing $x^{k+1} = x^k \cdot x$ and using the product rule, then rule (iii) and the induction hypothesis,

$$\Diamond x^{k+1} = x\,\Diamond x^k + x^k\,\Diamond x = x \cdot k x^{k-1} + x^k = (k+1)x^k,$$

so the result holds for $n = k + 1$. By induction it holds for all $n \geq 1$. For $n = 0$, $x^0 = 1$ is a constant, so $\Diamond x^0 = 0$ by the first part. In every case $\Diamond x^n = \frac{\mathrm{d}}{\mathrm{d}x} x^n$.

Step 2: sums. Rule (ii) is stated for two polynomials, but it extends to any finite sum by induction on the number of terms: if $\Diamond(\mathrm{f}_1 + \cdots + \mathrm{f}_m) = \Diamond \mathrm{f}_1 + \cdots + \Diamond \mathrm{f}_m$, then applying rule (ii) to $(\mathrm{f}_1 + \cdots + \mathrm{f}_m) + \mathrm{f}_{m+1}$ gives the same statement for $m + 1$ terms.

Step 3: putting it together. Any polynomial can be written $\mathrm{h}(x) = a_0 + a_1 x + \cdots + a_n x^n$ with constant coefficients a_k. By Step 2, then rule (iv) on each term, then Step 1,

$$\begin{aligned} \Diamond \mathrm{h}(x) &= \Diamond(a_0) + \Diamond(a_1 x) + \cdots + \Diamond(a_n x^n) \\ &= a_0\,\Diamond 1 + a_1\,\Diamond x + \cdots + a_n\,\Diamond x^n \\ &= 0 + a_1 + 2a_2 x + \cdots + n a_n x^{n-1} \\ &= \frac{\mathrm{d}}{\mathrm{d}x}\big(\mathrm{h}(x)\big), \end{aligned}$$

which is what we had to prove.

Checks

It's worth confirming that differentiation really does obey all four rules, otherwise the result would be vacuous; it does (the product rule, linearity and $\frac{\mathrm{d}}{\mathrm{d}x} x = 1$). We also built $\Diamond$ by computer using only the four rules, as in the solution, and compared it with the ordinary derivative on 500 random polynomials of degree up to 7 with fractional coefficients. They always agreed, and agreed with a numerical difference quotient; for example the rules give $\Diamond x^3 = 3x^2$.

Common slips

- Saying "$\Diamond c = 0$ because the derivative of a constant is zero". That assumes what we are trying to prove; the zero has to come from the rules.
- Forgetting the constant term in the final part. The induction only starts at $n = 1$, so a_0 needs the result for constants.
- Using rule (ii) for a sum of many terms without noting that it extends from two terms by induction.

Beyond STEP

An operation obeying rules (i), (ii) and (iv) is called a *derivation*, and derivations are studied throughout algebra and geometry. This question shows that on polynomials a derivation is fixed entirely by its value on x. More generally, $\Diamond \mathrm{h} = \mathrm{p}(x)\,\mathrm{h}'(x)$ where $\mathrm{p} = \Diamond x$, and in differential geometry this idea becomes the definition of a vector field. For the algebraic setting of polynomial rings, see Allenby, *Rings, Fields and Groups*.

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